D07 · Warehouse Operations & WMS

02 · WMS implementation: receiving, put-away & inventory control

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

A Warehouse Management System (WMS) directs and records every physical movement — receiving against ASN, directed put-away, replenishment, picking, cycle counting — so that system inventory matches physical inventory by location, lot and serial.

2 WHY IT MATTERS

Inventory accuracy is the foundation of every promise the company makes. **If the WMS does not know exactly what is where, ATP over-promises, pickers search, and safety stock grows to cover uncertainty.** The largest gains usually come from discipline the system enforces — scan-confirmed receiving and directed put-away — rather than from advanced features.

3 HOW IT IS DONE

1. **Receive against ASN with scanning.** Confirm quantity, lot and expiry at the dock; flag discrepancies immediately.
2. **Direct every put-away.** The system decides the location; operators confirm by scan.
3. **Replace annual counts with cycle counting.** ABC-based frequencies and root-cause analysis of every discrepancy.
4. **Integrate with the ERP in real time.** One inventory, not two; train every operator before go-live.

IN PRACTICE

WMS-ERP integration is the prerequisite of WMS value

A WMS disconnected from the ERP generates two versions of inventory: the financial system's and the warehouse's. Real-time integration ensures that inventory available for sale (in the ERP/OMS) exactly reflects the physical warehouse inventory.

Configure directed put-away from day 1 — don't defer it

Undirected put-away (the operator decides where to place product) is the most frequent root cause of low IA and warehouse disorder. Directed put-away configuration is the most impactful process change of the WMS implementation.

Wave planning and batch picking are the highest-impact productivity modules

Single-order picking generates maximum walking distance. Batch picking — collecting multiple orders in a single warehouse pass — can reduce walking distance 40–60% with a well-configured WMS.

Warehouse team training is 40% of implementation success

A WMS poorly used by the team is worse than no WMS — because the system records incorrect transactions that degrade IA and generate fulfillment errors. Hands-on training for all operators before go-live is non-negotiable.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: WMS implementation — pharmaceutical distribution DC, paper-based operation — Current operation processes 2,400 lines/day with 85 operators. IA is 91.4% and urgent order processing time is 4.2 hours.

WMS module implemented	Timeline	Measured impact
ASN receiving + scanner confirmation (vs. paper)	Month 1–2	Receiving errors: 18/month -> 1/month · Receiving time per load: -38%

WMS module implemented	Timeline	Measured impact
Directed put-away + FEFO rules (pharma)	Month 2–3	Product expiry from incorrect put-away: 0 incidents in 12 months post-go-live
Wave planning + batch picking for standard orders	Month 3–4	LPPH: 54 -> 78 lines/hr (+44%) · Operators required: 85 -> 68 (-20%)
Inventory accuracy management + daily cycle count	Month 4–5	IA: 91.4% -> 98.6% (+7.2 pp)

Result: 6-month total: LPPH +44%, IA +7.2 pp, urgent order time: 4.2 hr -> 1.8 hr. WMS ROI at 12 months: $1.4M in labor efficiency + $280K in error and expiry reduction.

HOW IT IS MEASURED

WMS Transaction Accuracy % (WMS transactions executed without error)

WMS Transaction Accuracy %(WMS transactions executed correctly without manual correction / Total transactions) × 100

Benchmark: >99.5% transaction accuracy in well-configured WMS processes A WMS accuracy <98% indicates transactions are being executed incorrectly with regularity — from master data problems, process errors, or omitted transactions.

Dock-to-Stock Time (time from truck arrival to product available in the system)

Dock-to-Stock TimeAverage time from receiving truck arrival to put-away confirmation and inventory availability in the WMS

Benchmark: <2 hours for standard receipts · <4 hours for high-complexity or quality-inspection receipts A long dock-to-stock time delays inventory availability for fulfillment — especially critical when the received product is needed to cover an urgent order.

WHAT YOU WOULD USE

WMS Platforms

SAP EWM (Extended Warehouse Management) — Module: WMS with Full SAP IntegrationThe reference WMS for SAP S/4HANA companies. Native integration with MM, SD, and PP. Supports multiple warehouse types and picking modalities.

Manhattan Associates WMS — Module: Best-in-Class WMS for Complex OperationsWorld-class WMS for high-complexity operations. Advanced labor management, slotting, and wave planning modules.

Blue Yonder Warehouse Management — Module: WMS + Labor Management + SlottingBlue Yonder WMS with advanced labor management and slotting optimization capabilities.

Mid-Market

Deposco / Extensiv (formerly 3PL Central) — Module: WMS for Mid-Market WarehousesWMS for mid-size warehouses with strong functionality-to-cost ratio.

4 SO WHAT

Implement the WMS for accuracy first. Scan-confirmed receiving, directed put-away, cycle counting and real-time ERP integration make inventory trustworthy — everything else builds on that.

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